

9. Orthogonal projections

In this chapter we compare the size of a subset of $\mathbf{R}^n$ with the sizes of its “shadows”. More precisely, we look for relations between the Hausdorff measures and dimension of the set and its orthogonal projections on $\gamma_{n,m}$ almost all m -dimensional linear subspaces $V \in G(n, m)$; recall Chapter 3 for the definitions and notations concerning $G(n, m)$. For the proofs we shall use capacities, and in terms of capacities also the results find sharper formulations. This study splits naturally into two parts according to whether for the dimension s of the given Borel set we have $s \leq m$ or $s > m$. In the first case the projections generically have the same dimension s while in the second they have positive $\mathcal{H}^m$ measure. It is clear that in some exceptional directions the dimension may decrease; for example for product sets. These basic results essentially go back to Marstrand [1]. At the end of the chapter we give some applications to self-similar sets and Brownian motion. The projections of integral dimensional sets will be studied in more details in Chapter 18.

Lipschitz maps and capacities

Note that for the orthogonal projections P_V , $V \in G(n, m)$, we have $\text{Lip}(P_V) \leq 1$. We have already seen in 7.5 that the Hausdorff measures and dimension cannot increase under such Lipschitz maps. We shall now prove the same for capacities. The following proof is due to Fuglede [1] simplifying the one given in Landkof [1, p. 158].

9.1. Theorem. *Let $f: A \rightarrow \mathbf{R}^m$, $A \subset \mathbf{R}^n$, be a Lipschitz map. Then*

$$C_s(fA) \leq \text{Lip}(f)^s C_s(A) \quad \text{for } s > 0.$$

Proof. We assume A , and so also fA , is compact. The general case can be reduced to this by approximation, which we leave to the reader; see Exercises 8.2 and 8.3. Let ν be a Radon measure with $\text{spt } \nu \subset fA$ and $\nu(fA) = 1$. By Theorem 1.20 there is a Radon measure μ such that $\text{spt } \mu \subset A$ and $f_\# \mu = \nu$. Then

$$1 = \nu(fA) = \mu(f^{-1}(fA)) = \mu(A),$$

whence by Theorem 1.19,

$$\begin{aligned}
 I_s(\nu) &= \iint |x - y|^{-s} df_{\#}\mu x df_{\#}\mu y \\
 &= \iint |f(u) - f(v)|^{-s} d\mu u d\mu v \\
 &\geq \text{Lip}(f)^{-s} \iint |u - v|^{-s} d\mu u d\mu v \\
 &= \text{Lip}(f)^{-s} I_s(\mu) \geq \text{Lip}(f)^{-s} C_s(A)^{-1}.
 \end{aligned}$$

Taking infimum over all such ν 's, we get

$$C_s(fA)^{-1} \geq \text{Lip}(f)^{-s} C_s(A)^{-1}.$$

□

Orthogonal projections, capacities and Hausdorff dimension

Before going into positive results in the other direction, we observe that for any dimensions n, m and s , $0 < s \leq m < n$, there are compact sets in $\mathbf{R}^n$ with positive $\mathcal{H}^s$ measure which are projected to zero $\mathcal{H}^s$ measure on all m -planes. We give an example only in the plane. It is taken from Martin and Mattila [1]. However, I do not know if there are such self-similar sets $K = \bigcup_{i=1}^N S_i(K)$ even when the different parts are disjoint and the similitudes S_i are composed only of translations and dilations.

9.2. Example. Given $0 < s \leq 1$ there is a compact set $F \subset \mathbf{R}^2$ such that $0 < \mathcal{H}^s(F) < \infty$ and $\mathcal{H}^s(P_L F) = 0$ for all $L \in G(2, 1)$.

Start from the disc $B_{0,0} = B(0, r_0)$ and take closed discs $B_{1,1}$ and $B_{1,2}$ inside it with disjoint interiors, their centres on the x -axis touching the boundary of $B_{0,0}$ and having radius r_1 with $2r_1^s = r_0^s$. Perform a similar operation inside both $B_{1,1}$ and $B_{1,2}$ but this time the new discs $B_{2,2i-1}$ and $B_{2,2i}$ inside $B_{1,i}$ lie on a diameter of $B_{1,i}$ making an angle α_1 with the x -axis. The new radius r_2 satisfies again $2r_2^s = r_1^s$. Continue this by turning at the k -th stage the direction of the diameters on which the new discs lie by angle α_k , see Figure 9.1. Let

$$F = \bigcap_{k=1}^{\infty} \bigcup_{i=1}^{2^k} B_{k,i}.$$

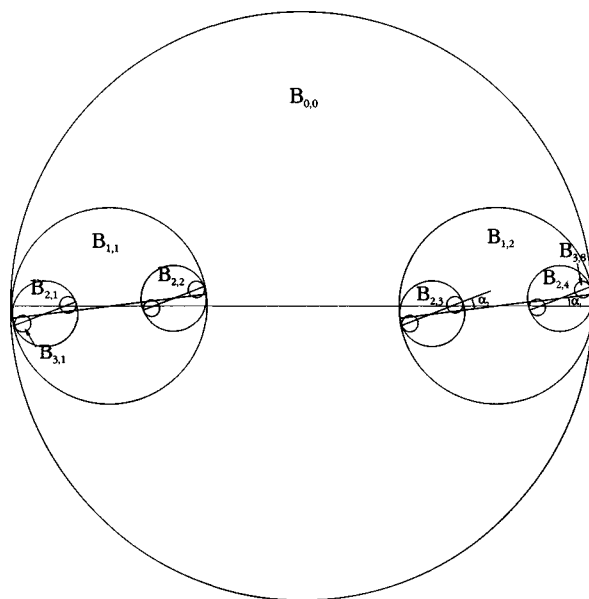


Figure 9.1.

Then by the methods of Chapter 4, $0 < \mathcal{H}^s(F) < \infty$. Select the angles α_k so that $\alpha_k \downarrow 0$ and $\sum_k \alpha_k = \infty$ and show that $\mathcal{H}^s(P_L F) = 0$ for $L \in G(2, 1)$ (or see Martin and Mattila [1]).

We shall now prove that the capacities behave much better under projections than the Hausdorff measures do. In the following m and n will be integers with $0 < m < n$.

9.3. Theorem. *Let $0 < s < m$. There is a constant c depending only on m, n and s such that for $A \subset \mathbb{R}^n$,*

$$\int^* C_s(P_V A)^{-1} d\gamma_{n,m} V \leq c C_s(A)^{-1}.$$

In particular, $C_s(A) > 0$ implies $C_s(P_V A) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$.

Proof. Let μ be a Radon measure with compact support such that $\text{spt } \mu \subset A$ and $\mu(A) = 1$. Then by Theorem 1.18 the image $P_{V\#}\mu$ is a Radon measure satisfying $\text{spt } P_{V\#}\mu \subset P_V A$ and $P_{V\#}\mu(P_V A) = 1$, consequently $C_s(P_V A)^{-1} \leq I_s(P_{V\#}\mu)$. Therefore by Theorem 1.19, Fubini's

theorem, and Corollary 3.12,

$$\begin{aligned} \int^* C_s(P_V A)^{-1} d\gamma_{n,m} V &\leq \int I_s(P_V \# \mu) d\gamma_{n,m} V \\ &= \iiint |P_V(x-y)|^{-s} d\mu x d\mu y d\gamma_{n,m} V \\ &= \iiint |P_V(x-y)|^{-s} d\gamma_{n,m} V d\mu x d\mu y \leq c I_s(\mu). \end{aligned}$$

Taking infimum over all such μ 's, we get the required inequality. $\square$

This leads immediately to the following corollary based on the relations between Hausdorff dimension and capacities, Theorem 8.9, and Theorem 7.5. It shows that although the Hausdorff measures may misbehave under projections, the dimension behaves properly. This simple approach via capacities was first used by Kaufman [1].

9.4. Corollary. *If $A \subset \mathbf{R}^n$ is a Borel set and $\dim A \leq m$, then $\dim P_V(A) = \dim A$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$.*

9.5. Remarks. We also have the inequalities

$$c^{-1} C_s(A) \leq \int_* C_s(P_V A) d\gamma_{n,m} V \leq C_s(A).$$

The right hand inequality follows immediately from Theorem 9.1. It is sufficient to verify the left hand inequality for compact sets A . It is a rather simple matter to show that the integrand is $\gamma_{n,m}$ measurable in this case, see Mattila [2].

We may assume $C_s(A) > 0$. Then $C_s(P_V A) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$ by Theorem 9.3. Hence Hölder's inequality and Theorem 9.3 give

$$\begin{aligned} 1 &= \left(\int C_s(P_V A)^{1/2} C_s(P_V A)^{-1/2} d\gamma_{n,m} V \right)^2 \\ &\leq \int C_s(P_V A) d\gamma_{n,m} V \int C_s(P_V A)^{-1} d\gamma_{n,m} V \\ &\leq c C_s(A)^{-1} \int C_s(P_V A) d\gamma_{n,m} V. \end{aligned}$$

We are now going to turn to the case $s > m$. We shall first study the behaviour of the projected measures $P_V \# \mu$ for a Radon measure μ on

$\mathbf{R}^n$ and then apply these results to capacities and Hausdorff dimension. For these applications we need to show that if $I_m(\mu) < \infty$ then for any A with $\mu(A) > 0$, $\mathcal{H}^m(P_V A) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$. But in some later applications we are going to need a bit more; namely, that the exceptional set of V 's can be taken to be independent of A . Looking more closely at the definitions, one sees that this means exactly that $P_{V\#}\mu$ is absolutely continuous with respect to $\mathcal{H}^m$. Actually we shall also obtain that the Radon–Nikodym derivative

$$\begin{aligned} (9.6) \quad D(P_{V\#}\mu, u) &\equiv D(P_{V\#}\mu, \mathcal{H}^m \llcorner V, u) \\ &= \lim_{\delta \downarrow 0} (2\delta)^{-m} P_{V\#}\mu(B(u, \delta)) \\ &= \lim_{\delta \downarrow 0} (2\delta)^{-m} \mu(\{x : |u - P_V x| \leq \delta\}) \end{aligned}$$

is L^2 -integrable with respect to $\mathcal{H}^m$ on V for $\gamma_{n,m}$ almost all V . Such results could be obtained with the aid of Fourier transform, see Kaufman [1] and Falconer [4, § 6.3], but we shall employ the differentiation theory of Chapter 2.

9.7. Theorem. *Let μ be a Radon measure on $\mathbf{R}^n$ with compact support and with $I_m(\mu) < \infty$. Then $P_{V\#}\mu \ll \mathcal{H}^m$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$ and*

$$\iint_V D(P_{V\#}\mu, u)^2 d\mathcal{H}^m u d\gamma_{n,m} V \leq c I_m(\mu),$$

where c is a constant depending only on n and m .

Proof. By Theorem 1.18 $P_{V\#}\mu$ is a Radon measure with compact support for $V \in G(n, m)$. Using Fatou's lemma, the definition of $P_{V\#}\mu$, Theorem 1.19, Fubini's theorem and Lemma 3.11, we compute, with $\underline{D}(P_{V\#}\mu, u) = \underline{D}(P_{V\#}\mu, \mathcal{H}^m \llcorner V, u)$,

$$\begin{aligned} &\iint \underline{D}(P_{V\#}\mu, u) dP_{V\#}\mu u d\gamma_{n,m} V \\ &\leq \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint P_{V\#}\mu(B(u, \delta)) dP_{V\#}\mu u d\gamma_{n,m} V \\ &= \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint \mu(\{y : |P_V(x - y)| \leq \delta\}) d\mu x d\gamma_{n,m} V \\ &= \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint \gamma_{n,m}(\{V : |P_V(x - y)| \leq \delta\}) d\mu x d\mu y \\ &\leq c I_m(\mu). \end{aligned}$$

Thus for $\gamma_{n,m}$ almost all $V \in G(n, m)$, $\underline{D}(P_{V\#}\mu, u) < \infty$ for $P_{V\#}\mu$ almost all $u \in V$, which by Theorem 2.12(3) means $P_{V\#}\mu \ll \mathcal{H}^m$, and so, by Theorem 2.12(1), $D(P_{V\#}\mu, u)$ exists for $P_{V\#}\mu$ almost all $u \in V$. Since $P_{V\#}\mu \ll \mathcal{H}^m$ implies

$$\int_V D(P_{V\#}\mu, u)^2 d\mathcal{H}^m u = \int D(P_{V\#}\mu, u) dP_{V\#}\mu u,$$

see Exercise 2.6, the theorem follows. $\square$

9.8. Corollary. *If $A \subset \mathbf{R}^n$ with $C_m(A) > 0$, then $\mathcal{H}^m(P_V A) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$.*

(Recall that $C_m(A) > 0$ for any Borel set A with $\dim A > m$ by Theorem 8.9.)

Proof. As $C_m(A) > 0$ there is a Radon measure μ with $\text{spt } \mu \subset A$, $\mu(A) = 1$, and $I_m(\mu) < \infty$. Then $P_{V\#}\mu(P_V A) = 1$, which gives $\mathcal{H}^m(P_V A) > 0$ for all V such that $P_{V\#}\mu \ll \mathcal{H}^m$. $\square$

If $s > m$, the condition $C_s(A) > 0$ is stronger than $C_m(A) > 0$, and it should lead to stronger consequences. Such stronger consequences will be given by measuring the size of the sections $P_V^{-1}\{u\} \cap A$, $u \in V$, in the following chapter.

Next we study capacities and projections in the case $s = m$.

9.9. Theorem. *Let μ be a Radon measure on $\mathbf{R}^n$ with $\mu(\mathbf{R}^n) = 1$ and $F = \text{spt } \mu$. Then*

$$\int \mathcal{H}^m(P_V F)^{-1} d\gamma_{n,m} V \leq c I_m(\mu),$$

where c is a constant depending only on m and n .

Proof. Since F is closed, an easy argument shows that $V \mapsto \mathcal{H}^m(P_V F)^{-1}$ is a Borel function. We may assume F is compact and $I_m(\mu) < \infty$. Then $P_{V\#}\mu \ll \mathcal{H}^m$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$ by Theorem 9.7. For any such V we obtain by Theorem 2.12(2) and Hölder's inequality

$$\begin{aligned} 1 &= P_{V\#}\mu(P_V F)^2 = \left(\int_{P_V F} D(P_{V\#}\mu, u) d\mathcal{H}^m u \right)^2 \\ &\leq \mathcal{H}^m(P_V F) \int_V D(P_{V\#}\mu, u)^2 d\mathcal{H}^m u. \end{aligned}$$

The required inequality follows now from Theorem 9.7. $\square$

9.10. Corollary. For any $A \subset \mathbf{R}^n$,

$$\int^* \mathcal{H}^m(P_V A)^{-1} d\gamma_{n,m} V \leq c C_m(A)^{-1}$$

and

$$C_m(A) \leq c \int_* \mathcal{H}^m(P_V A) d\gamma_{n,m} V.$$

The first inequality follows directly from Theorem 9.9 and the definition of capacities; see Exercise 8.2. The second follows from the first by Hölder's inequality as in 9.5.

9.11. Remarks. The last results 9.9 and 9.10 are from Mattila [15]. There also some sharp constants were obtained. The inequalities of Theorems 9.9 and 9.10 were proved in Mattila [15] with constants which give equality for balls for $m = n - 2$ and $m = n - 1$, but quite likely not for $m < n - 2$. For example when $n = 2$ and $m = 1$, $c = 1/\pi$ in 9.9 and 9.10. This also leads to some precise inequalities relating Lebesgue measure and orthogonal projections. Namely, the classical capacity C_{n-2} , $n \geq 3$, can also be obtained through a minimization of Dirichlet's integrals, recall 8.12 (2), which yields, see Mattila [15],

$$\mathcal{L}^n(A)^{m/n} \leq c_1(n, m) C_m(A), \quad A \subset \mathbf{R}^n,$$

for $m = n - 2 \geq 1$ with equality for balls. Combining this with the aforementioned sharp inequalities between capacities and orthogonal projections, we obtain for $m = n - 2 \geq 1$,

$$(1) \quad \mathcal{L}^n(A)^{m/n} \leq c_2(n, m) \int_* \mathcal{H}^m(P_V A) d\gamma_{n,m} V$$

with equality for balls. This holds also for $m = 1$, and it was proved by M. Chlebík (unpublished). He worked with finite unions of convex sets and not with capacities. For other values of m such optimal inequalities seem to be unknown. For convex sets they are well-known, see Burago and Zalgaller [1].

If (1) holds for $m = n - 1$ in this sharp form, as it does when $n = 2$, it leads to the classical isoperimetric inequality. For example, if A is a bounded open set with smooth boundary, one can show by standard integral geometry that

$$\int \mathcal{H}^{n-1}(P_V A) d\gamma_{n,n-1} V \leq c_3(n) \mathcal{H}^{n-1}(\partial A)$$

with equality for convex sets. In fact by approximation this generalizes to any bounded Borel set A provided $\mathcal{H}^{n-1}(\partial A)$ is replaced by the perimeter $P(A)$ of A in the distributional sense, see Giusti [1], L. Simon [1] or Ziemer [1]. Combining this with (1) we get the isoperimetric inequality

$$\mathcal{L}^n(A)^{(n-1)/n} \leq c_4(n) P(A)$$

with equality for balls.

9.12. Further remarks. (1) We briefly discuss some other sharpenings for the results concerning projections and dimension. Let us consider only sets in $\mathbf{R}^2$. For the higher dimensional variants see Mattila [2], Kaufman and Mattila [1] and Falconer [3].

Kaufman [1] proved that if A is a Borel set in $\mathbf{R}^2$ with $\dim A = s < 1$, then

$$\mathcal{H}^s(\{L \in G(2, 1) : \dim P_L A < s\}) = 0.$$

This is sharp in the sense that for each s , $0 \leq s \leq 1$, there exists a Borel set $A \subset \mathbf{R}^2$ such that $\dim A = s$ and

$$\dim\{L \in G(2, 1) : \dim P_L A < s\} = s,$$

see Kaufman and Mattila [1] or Falconer [4, Theorem 8.17]; the example makes use of number-theoretic results on Hausdorff dimension, see also Kaufman [2]. The proof of Kaufman's result is similar to that given above for Theorem 9.3: replace $\gamma_{2,1}$ with a measure ν on $G(2, 1)$ satisfying $\nu(B(L, \delta)) \leq c\delta^s$, and use Frostman's lemma, Theorem 8.8. Kenyon [2] studied the exceptional set of projections for the Sierpinski gasket (recall 5.1), see also Peres [2].

In the case $1 < s < 2$, Falconer [3] proved for Borel sets $A \subset \mathbf{R}^2$ with $C_s(A) > 0$,

$$\mathcal{H}^{2-s}(\{L \in G(2, 1) : \mathcal{H}^1(P_L A) = 0\}) = 0,$$

and $2 - s$ is again the best possible dimension. Falconer also derived an upper bound (which for $n = 2$, $m = 1$ and $\dim A > 1$ is $1 + r - \dim A$) for

$$\dim\{V \in G(n, m) : \dim P_V A < r\}$$

for a Borel set $A \subset \mathbf{R}^n$ and $0 < r \leq \dim A$. It is not known if it is sharp in all cases. Falconer's proofs make use of the Fourier transform.

(2) Dekking and Grimmett [1] and Falconer [14] obtained results for the projections of some random Cantor sets in a fixed direction. The

problem of determining the dimension of the horizon of landscape, see Falconer [17], has also a projection flavour.

(3) Davies [5] showed that the results of this chapter are not true for arbitrary sets without measurability assumptions. His constructions are based on the continuum hypothesis.

(4) Järvenpää [1] has shown that the analogue of Corollary 9.4 is not valid for the upper Minkowski and packing dimensions. She also obtained a lower bound, but Falconer and Howroyd [1] proved the sharp inequality: let Dim denote any of the dimensions $\overline{\text{dim}}_M$, $\underline{\text{dim}}_M$, $\overline{\text{dim}}_p$ or $\underline{\text{dim}}_p$. Then for Borel sets $A \subset \mathbf{R}^n$,

$$\text{Dim } P_V(A) \geq \frac{\text{Dim } A}{1 + (1/m - 1/n) \text{Dim } A}$$

for $\gamma_{n,m}$ almost all $V \in G(n, m)$.

Self-similar sets with overlap

We now present some applications of the projection theorems. The first is on “generic” self-similar sets. Recall from 4.13 that for any finite sequence $S_1, \dots, S_N$, $N \geq 2$, of similitudes of $\mathbf{R}^n$ there corresponds a unique non-empty compact invariant set K such that

$$K = \bigcup_{i=1}^N S_i K.$$

If the open set condition is satisfied, the Hausdorff dimension s of K is given by $\sum_{i=1}^N r_i^s = 1$ where $r_i = \text{Lip}(S_i)$ is the contraction ratio of S_i . The projection theorem 9.4 can be used to show that often even without the open set condition $\text{dim } K$ is given by the same formula for “generic” similitudes $S_1, \dots, S_N$. The following result was proved by Falconer [11].

9.13. Theorem. *Let $T_i: \mathbf{R} \rightarrow \mathbf{R}$, $i = 1, \dots, N$, $N \geq 2$, be linear similitudes given by $T_i x = \lambda_i x$ where $\lambda_i \neq 0$ and $\sum_{i=1}^N |\lambda_i| < 1$. Then for $\mathcal{L}^N$ almost all $(c_1, \dots, c_N) \in \mathbf{R}^N$ the non-empty compact invariant set K ,*

$$K = \bigcup_{i=1}^N (T_i + c_i)(K),$$

has Hausdorff dimension s where $\sum_{i=1}^N |\lambda_i|^s = 1$.

Proof. The idea is to lift the mappings T_i to similitudes $S_i: \mathbf{R}^N \rightarrow \mathbf{R}^N$ which satisfy the open set condition, apply the projection theorem 9.4 to the invariant set of the similitudes S_i and arrange the situation so that the statement thus obtained corresponds to the statement desired.

The similitudes $S_i: \mathbf{R}^N \rightarrow \mathbf{R}^N$, $i = 1, \dots, N$, are defined by

$$S_i(x, y) = (\lambda_i x, \lambda_i y + a_i), \quad x \in \mathbf{R}, \quad y \in \mathbf{R}^{N-1},$$

where the vectors $a_i \in \mathbf{R}^{N-1}$ are chosen so that for $Q = \{x \in \mathbf{R}^N : |x_i| < 1 \text{ for } i = 1, \dots, N\}$,

$$S_i Q \subset Q \quad \text{and} \quad S_i Q \cap S_j Q = \emptyset \quad \text{for } i \neq j.$$

This is easily arranged. Furthermore, by a slight perturbation, we may find the points a_i so that the vectors $(a_i, 1 - \lambda_i) \in \mathbf{R}^N$, $i = 1, \dots, N$, span $\mathbf{R}^N$. By Theorem 4.14 there exists a unique non-empty compact set H such that

$$(1) \quad H = \bigcup_{i=1}^N S_i H \quad \text{and} \quad \dim H = s.$$

For $t \in \mathbf{R}^{N-1}$ define $p_t: \mathbf{R}^N \rightarrow \mathbf{R}$ by

$$p_t(x, y) = x + t \cdot y, \quad x \in \mathbf{R}, \quad y \in \mathbf{R}^{N-1}.$$

These mappings are essentially projections and we can apply Corollary 9.4 to them (by either transforming the maps p_t to projections or checking that the proof of 9.4 works directly to them). Thus by (1) we obtain

$$(2) \quad \dim p_t(H) = s \quad \text{for } \mathcal{L}^{N-1} \text{ almost all } t \in \mathbf{R}^{N-1}.$$

Since

$$(T_i + a_i \cdot t)(p_t(x, y)) = \lambda_i x + \lambda_i t \cdot y + a_i \cdot t = p_t(S_i(x, y)),$$

we have by (1),

$$p_t(H) = \bigcup_{i=1}^N (p_t \circ S_i)(H) = \bigcup_{i=1}^N (T_i + a_i \cdot t)(p_t(H)),$$

so that $p_t(H)$ is the unique non-empty compact invariant set associated with the similitudes $T_i + a_i \cdot t$, $i = 1, \dots, N$. It follows that for any $u \in \mathbf{R}$, $p_t(H) + u$ is the invariant set associated with $T_i + a_i \cdot t + (1 - \lambda_i)u$, $i = 1, \dots, N$, and, by (2), $\dim(p_t(H) + u) = s$ for $\mathcal{L}^{N-1}$ almost all $t \in \mathbf{R}^{N-1}$. This proves the theorem since the mapping $(t, u) \mapsto (a_1 \cdot t + (1 - \lambda_1)u, \dots, a_N \cdot t + (1 - \lambda_N)u)$ is a linear bijection of $\mathbf{R}^N$; in particular it is onto and it preserves the sets of $\mathcal{L}^N$ measure zero. $\square$

9.14. Remarks. The above proof generalizes to some higher dimensional cases but it is not clear if this idea can be used to treat the general case of linear similitudes $T_i: \mathbf{R}^n \rightarrow \mathbf{R}^n$. On the other hand, the method can also be modified to apply to some attractors related to non-linear mappings, see Falconer [11]. For similar results on the dimension of generic self-affine sets, see Falconer [12] and Falconer and Marsh [1]. For other applications of the results and methods of this chapter to invariant sets or related concepts, see Kenyon and Peres [2], Ledrappier [1] and Ledrappier and Young [1].

Brownian motion

Next we give some basic facts about Brownian motion. We shall be rather brief, for more details, see Falconer [4], [16] and Kahane [3], and the references given there. A survey on dimensional properties of various sets related to Brownian motion and other stochastic processes is given in Taylor [2].

9.15. Brownian motion. The n -dimensional Brownian motion is a probability measure P_n on the space Ω_n of continuous functions $\omega: [0, \infty) \rightarrow \mathbf{R}^n$ with $\omega(0) = 0$ such that the increments $\omega(t_2) - \omega(t_1)$ and $\omega(t_4) - \omega(t_3)$ are independent for $0 \leq t_1 \leq t_2 \leq t_3 \leq t_4$ and such that $\omega(t+h) - \omega(t)$ has Gaussian distribution with zero mean and variance h for $t \geq 0$ and $h > 0$. In particular,

$$P_n(\{\omega : |\omega(t+h) - \omega(t)| \leq \varrho\}) = ch^{-n/2} \int_0^\varrho r^{n-1} e^{-r^2/(2h)} dr$$

for $t \geq 0$, $h > 0$ and $\varrho > 0$, where c is a positive constant. From this one gets easily with the help of Theorem 1.15 (forgetting here and below the not so simple measurability questions)

$$(9.16) \quad \int |\omega(t+h) - \omega(t)|^{-s} dP_n \omega = c_1 h^{-s/2}$$

for $t \geq 0$, $h > 0$ and $0 < s < n$.

9.17. Lemma. *Let $0 < s < 1$, with $0 < s < 1/2$ if $n = 1$. If $A \subset [0, \infty)$ and $C_s(A) > 0$, then $C_{2s}(\omega(A)) > 0$ for P_n almost all $\omega \in \Omega_n$.*

Proof. By the definition of C_s there is a Radon measure $\mu \in \mathcal{M}(A)$ with $I_s(\mu) < \infty$. Then by Theorem 1.19, Fubini's theorem and (9.16),

$$\begin{aligned} \int I_{2s}(\omega_{\#}\mu) dP_n\omega &= \iiint |\omega(t) - \omega(u)|^{-2s} d\mu t d\mu u dP_n\omega \\ &= c_1 I_s(\mu) < \infty. \end{aligned}$$

Thus $I_{2s}(\omega_{\#}\mu) < \infty$ for P_n almost all ω . Since $\text{spt}(\omega_{\#}\mu) \subset \omega(A)$, we obtain $C_{2s}(\omega(A)) > 0$ for P_n almost all $\omega \in \Omega_n$. $\square$

The above lemma combined with Theorem 8.9 gives the almost sure lower bound $2s, \min\{2s, 1\}$ in the case $n = 1$, for the Hausdorff dimension of $\omega(A)$ when A is a Borel set with $\dim A = s$. That this is also an upper bound follows from the fact that P_n almost every ω is Hölder continuous with exponent λ for $0 < \lambda < 1/2$, see Falconer [4], [16] or Kahane [3]. Thus we have

9.18. Theorem. *Let $A \subset [0, \infty)$ be a Borel set. Then for P_n almost all $\omega \in \Omega_n$,*

$$\begin{aligned} \dim \omega(A) &= 2 \dim A && \text{if } n \geq 2 \text{ or } \dim A \leq 1/2, \\ \dim \omega(A) &= 1 && \text{if } n = 1 \text{ and } \dim A > 1/2. \end{aligned}$$

In fact, the last statement can be sharpened to $\mathcal{L}^1(\omega(A)) > 0$ and we shall prove this under the weaker assumption $C_{1/2}(A) > 0$. A proof based on the Fourier transform can be found in Kahane [3]. The method below applying the projection theorem 9.3 was observed by Pavel Etinghoff and Yuval Peres. Under the assumption $\dim A > 1/2$ Kaufman [4] proved that $\omega(A)$ has even non-empty interior P_1 almost surely.

9.19. Theorem. *Let $A \subset [0, \infty)$ with $C_{1/2}(A) > 0$. Then $\mathcal{L}^1(\omega(A)) > 0$ for P_1 almost all $\omega \in \Omega_1$.*

Proof. By Lemma 9.17 $C_1(\omega(A)) > 0$ for P_2 almost all $\omega: [0, \infty) \rightarrow \mathbf{R}^2$. Hence Corollary 9.8 yields $\mathcal{L}^1(p_\theta(\omega(A))) > 0$ for $\mathcal{H}^1$ almost all $\theta \in S^1$ where $p_\theta: \mathbf{R}^2 \rightarrow \mathbf{R}$ is given by $p_\theta x = \theta \cdot x$ for $\theta \in S^1$. It follows that for $\mathcal{H}^1$ almost all $\theta \in S^1$, $\mathcal{L}^1((p_\theta \circ \omega)(A)) > 0$ for P_2 almost all $\omega \in \Omega_2$. But $\{p_\theta \circ \omega: \omega \in \Omega_2\}$ equipped with the measure P_2 provides a model

for the one-dimensional Brownian motion and the theorem follows from the uniqueness of such a model, see Falconer [16, § 16.1]. $\square$

9.20. Remarks. (1) Using the quantitative estimate of 9.10 one gets a quantitative lower bound for $\mathcal{L}^1(\omega(A))$ in 9.19. For other results of a similar type, see Kahane [3], Kaufman [3], [6] and [7].

(2) Applications of projection theorems to convex sets can be found in Dalla and Larman [1], see also Falconer [4, § 8.6], and to curve packing problems (see Remark 18.13 (1)) in Falconer [3] and [4, § 7].

Exercises.

1. Show that $\dim P_V A \geq \dim A + m - n$ for all $A \subset \mathbf{R}^n$, $V \in G(n, m)$.
2. Define $f_t: \mathbf{R}^2 \rightarrow \mathbf{R}$ by $f_t(x, y) = x + ty$. Show that for Borel sets $A \subset \mathbf{R}^2$ with $\dim A \leq 1$, $\dim f_t(A) = \dim A$ for $\mathcal{L}^1$ almost all $t \in \mathbf{R}$.
3. Let $E = C(\lambda) \times C(\lambda)$, $0 < \lambda \leq 1/4$, recall 4.10, and let $s = \log 4 / \log(1/\lambda)$ so that $0 < \mathcal{H}^s(E) < \infty$. Find countably many lines $L \in G(2, 1)$ such that $\mathcal{H}^s(P_L E) = 0$. (Deciding whether there can be uncountably many does not seem to be easy.)
4. Let E and s be as in the preceding exercise with $0 < \lambda < 1/4$. Show that there is a non-empty open set $G \subset G(2, 1)$ such that $\mathcal{H}^s(P_L E) > 0$ for $L \in G$. Hint: Look for lines L for which $P_L|_E$ has Lipschitz inverse. (I don't know if $\mathcal{H}^s(P_L E) > 0$ for $\gamma_{2,1}$ almost all $L \in G(2, 1)$ except for small λ 's; see the next exercise.)
5. Let E and s be as above with $2 \dim E = 4 \log 2 / \log(1/\lambda) < 1$. Show that $\mathcal{H}^s(P_L E) > 0$ for $\gamma_{2,1}$ almost all $L \in G(2, 1)$. Hint: This idea is due to Marstrand. Conclude from Corollary 8.11 that $\mathcal{H}^1(\{x - y : x, y \in E\}) = 0$ and show that $\mathcal{H}^s(P_L E) > 0$ if the line L is not orthogonal to any vector $x - y$ with $x \neq y$, $x, y \in E$.